

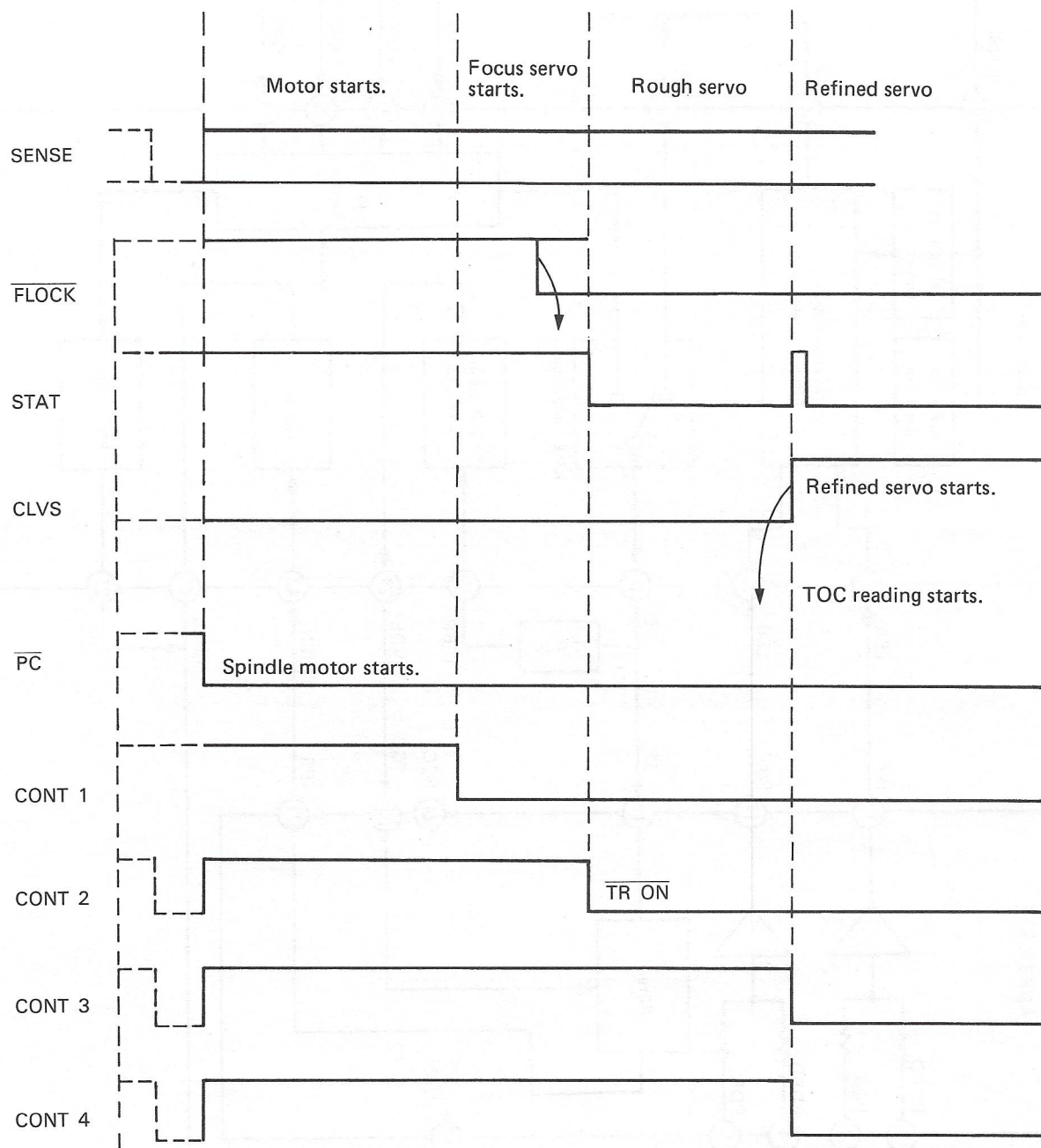
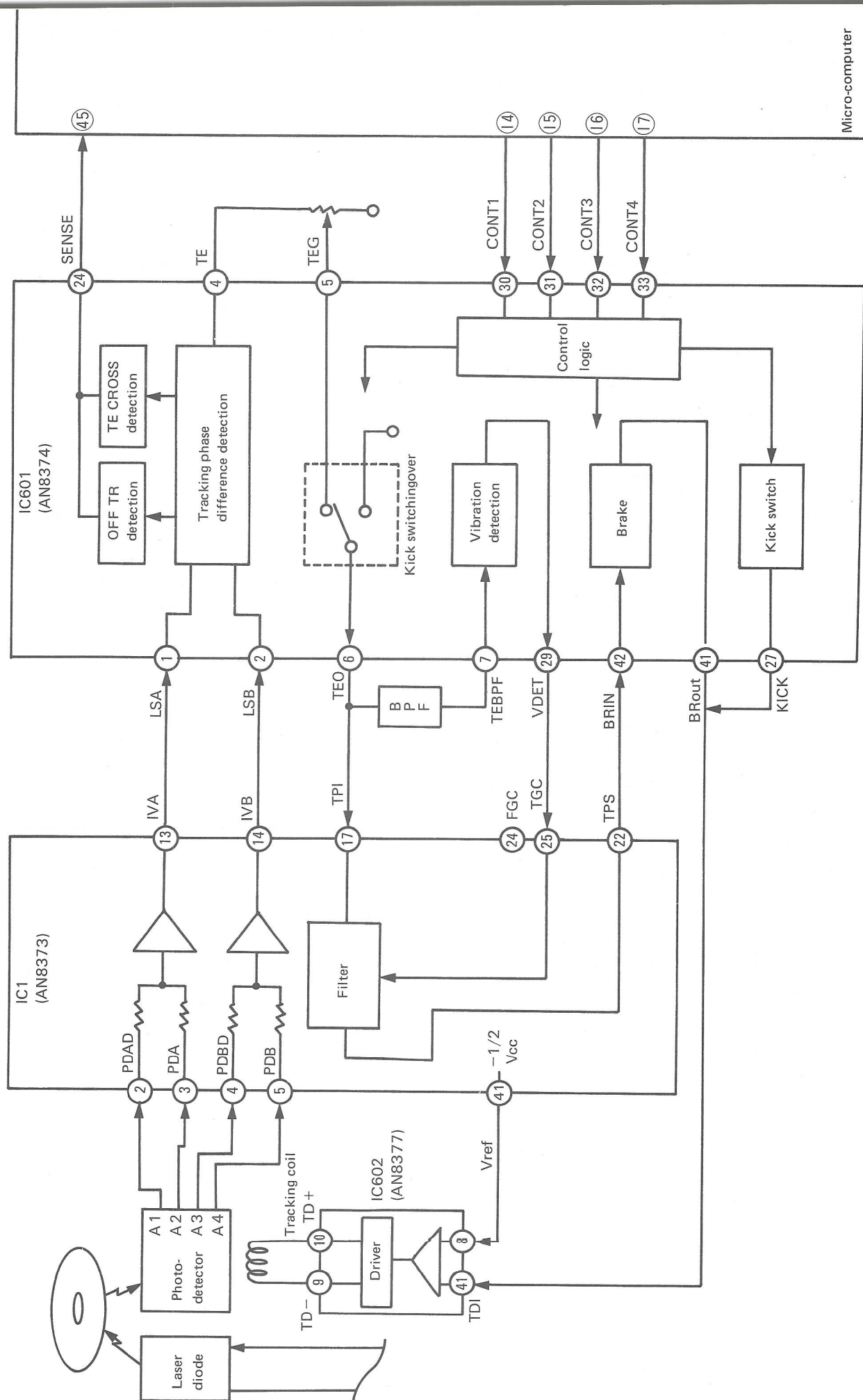


Fig. 4-5 Spindle Motor Timing Chart from Starting to FOC Read-in

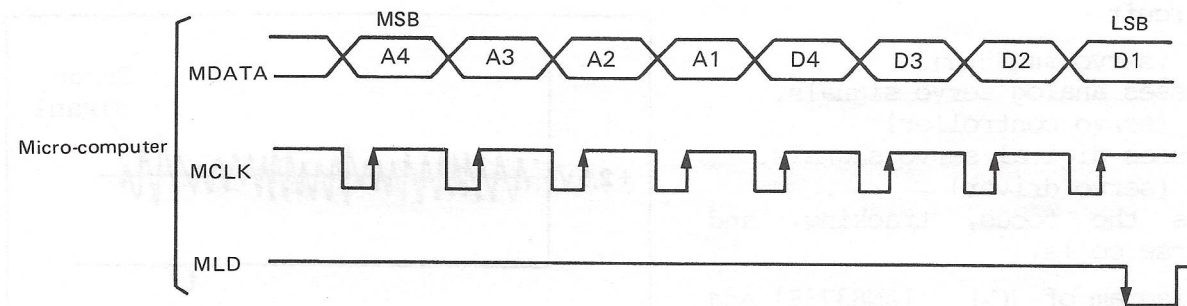
Fig. 4-6 Tracking Servo Circuit



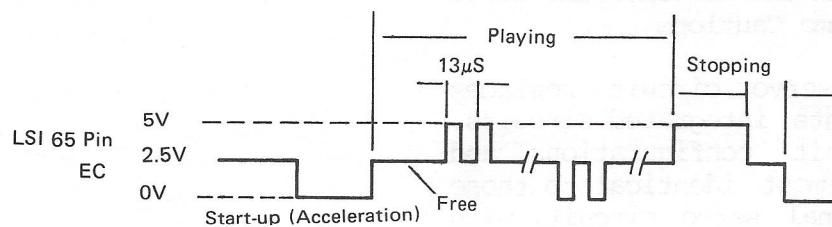
(4) Timing Chart

No.	←(MSB) 8 bits (LSB)→								Remarks
	A4	A3	A2	A1	D4	D3	D2	D1	
①	0	0	0	1	1 (with FG)	1 (TT ON)	0	0	CLV mode
							0	1	
					0 (without FG)	0 (TT OFF)	1	0	
							1	1	
②	0	0	1	0	—	—	1 (ATT-12dB)	1 (MUTE ON)	Interpolation circuit
							0 (ATT OFF)	0 (MUTE OFF)	
③	0	1	1	1	—	—	0	0	Status output switching
							1 (CLVOK)	0	
							1 (TT STOP)	1	

Contents of M. DATA



Transferring M. DATA (Micro-computer → LSI)



CLV Control Output of EC Terminal

A4-4 Description of New Servo Circuit (MN8373S and MN8374S)

1. General

The servo circuit in CQ-E850, CQ-DP25/26 has been entirely changed with new IC's for performance improvement and compactness.

2. Circuit Configuration

The servo circuit used in the conventional FF-1 CD players comprised a servo IC (AN8370S) and discrete circuit. In contrast, the modified servo circuit consists of three IC chips with on discrete circuit.

- (1). IC-1 (servo amplifier)
=Processes analog servo signals.
- (2). IC-601 (servo controller)
=Processes digital servo signals.
- (3). IC-602 (servo driver)
=Drives the focus, tracking, and traverse coils.

The block diagram of IC-1 (AN8373S) and IC-601 (AN8374S) are shown in figure 4-9.

3. Departures from the Conventional Servo Circuit and some Cautions

While the new servo circuit replaces discrete components integrated circuits, the basic circuit configurations and operations are almost identical to those of the conventional servo circuit, with the exception of the following details:

- (1). PLL adjustment eliminated
=Servo signals are automatically pulled into the PLL lock range by using the CLK (88.2KHz) signal.
- (2). 2.5V reference for tracking error and focus error signals
=Because the new circuit operates on a signal +5V supply, the reference voltage for the error signals has been changed to 2.5V as shown in figure 4-7.

NOTE

The 2.5V reference is also used for turntable height adjustment.

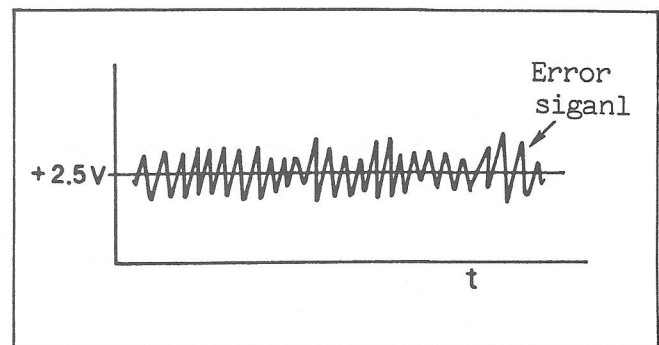


Figure 4-7

Focus and Tracking Error Signal

5. Data Signal Flow

(1). RF Signal Flow

- * The photo detectors A1 - A6 output is applied to IC-101's RF IN INPUT (pin 39) via an RF amplifier within the Optical Pick-up.
- * After being amplified by another RF amplifier within IC-101, the signal passes through an equalizer for 3T compensation, goes into the AGC (Auto Gain Control) circuit where it undergoes automatic gain control, and appears at the ARF (RF signal controlled by auto gain control) output (pin 35) of IC-101.

(2). EFM Data Generation

- * The RF signal applied to IC-102's ARF input (pin 12) is sliced by the DSL (data sliced) into a pulse signal EFM (eight to fourteen bit modulation), and then coupled to the PLL circuit.
- * The PLL is synchronized with the EFM signal and then coupled to the signal processing block.

6. PLL Operations

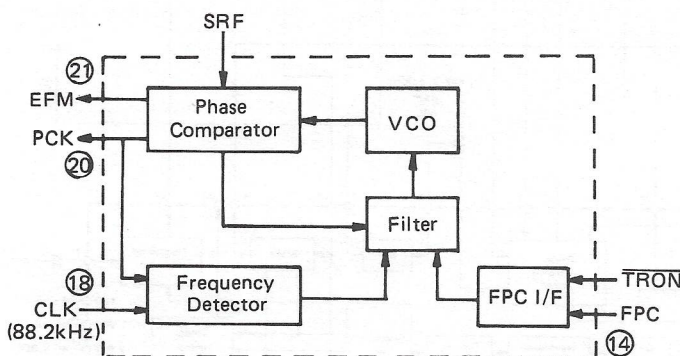


Figure 4-8 PLL Block

- (1). The new PLL circuit has replaced the PLL adjustment with a frequency comparator, one of whose input is clocked by the CLK signal (88.2KHz/11.34us) supplied from the digital signal processor.

The PLL circuit shows figure 4-8.

- (2). If PLL lock is lost, the frequency comparator's error output, this error signal will be controlled VCO's free-running frequency. The large error output of the frequency comparator drives the VCO until range. When the PLL is locked (TRON active), the FPC (frequency comparison error signal) is active and the VCO is controlled by the phase comparator's error output.

7. Control Logic (IC-601)

Commands used to set the servo circuit status are supplied from the system CPU in parallel 4 bit data (CNT1 - CNT4).

SENSE is the status read signal for IC-601, which is transferred to the system CPU. All other control logic operations are identical to those of the conventional control logic.

8. Focus Servo Circuit Operation

- (1). The outputs of the photo detector are coupled to pin 2 - 5 of IC-101, where they are phase corrected by the A1 + A2 and A3 + A4 signals and converted from current to voltage signals. They are transferred to the focus balance circuit and subjected to photo detector position error correction by means of VR101 (best eye adjustment).

- (2). The difference between the A1 + A2 and A3 + A4 signals, produced within the FE mixing circuit, serves as the Focus Error signal which appears at IC-101 pin 15.

The signal is then applied to IC-601 pin 8. VR63 is used for offset voltage injection control.

- (3). Since the Focus switch is closed in the PLAY mode, the FE signal appears at IC-601 pin 9 and couples to the loop filter within IC-101, where it is used to determine the focus servo's frequency response. The output of the loop filter at IC-101 pin 23, is transferred to the focus driving circuit and drives the focus actuator.

4. Servo Circuit Block Diagram

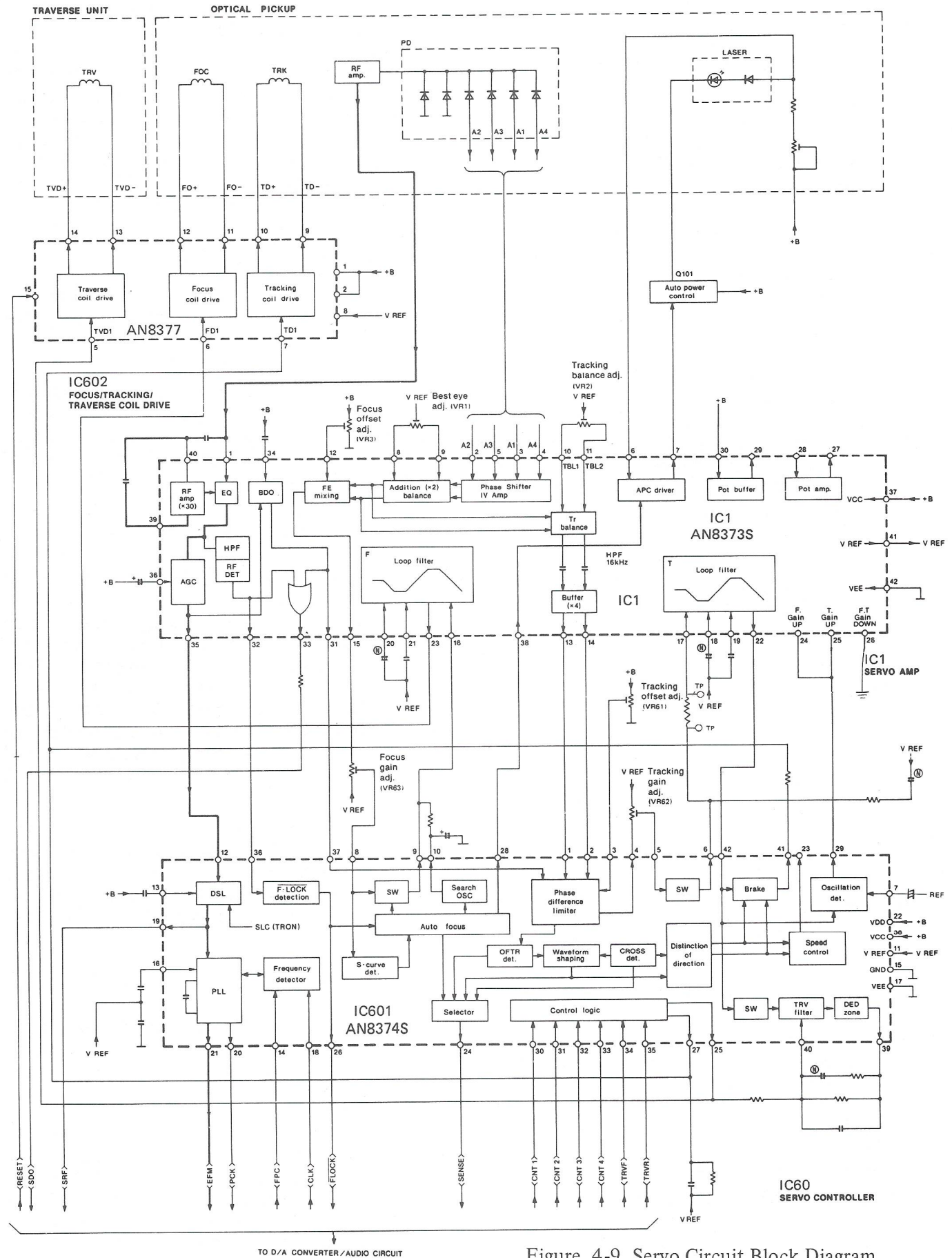


Figure 4-9 Servo Circuit Block Diagram

9. Tracking Servo Circuit Operation

- (1). The focus balance circuit output, $A1 + A2$ and $A3 + A4$, are also transferred to the tracking balance circuit. The output of the tracking balance circuit are applied to the phase difference limiter within IC-601, where they are phase compared to produce a tracking error signal.

The tracking error signal at IC-601 pin 4 is subjected to gain adjustment, then applied to the loop filter, within IC-101 via the tracking switch.

- (2). The loop filter output appearing at IC-601 pin 22, is coupled to pin 42 (BR IN/ tracking error input) of IC-601, where it is fed into a braking circuit. The output of the braking circuit appears at pin 41 (BR OUT/ tracking drive control signal) and couples to the tracking coil driver to drive the tracking actuator's.

10. Traverse Servo Circuit Operation

- (1). PLAY mode:

The tracking error signal applied to IC-601 pin 42 goes through the traverse switch and TRV (traverse circuit) filter (for response compensation) and appears at pin 39. This output is used to control the traverse actuator.

- (2). KICK mode:

The KICK signal decoded from the system CPU command by the control logic appears at pin 27. It is used to control the tracking actuator which KICKs the pick-up by a span of 1, 10, or 100 tracks according to the system CPU command.

- (3). Forced traverse mode:

The TRVF, TRVR signals supplied from system CPU to pins 34 and 35, respectively, are decoded to produce a traverse drive signal. The signal appears at pin 25 and drives the traverse coil.

11. RF Signal Detection and Auto Focus Circuit

The auto focus circuit within IC-601 performs focus search. If it detects an FE signal S-Curve it closes the focus switch contacts to close the focus servo loop. At this point the RF detector within IC-101 detects the RF signal and supplies it to the F.LOCK DET. input at IC-601 pin 36.

The F.LOCK signal is used to indicate to system CPU that focus is "IN". If focus goes "OUT" and the RF signal is no longer detected, the auto focus circuit opens the focus loop switch and starts focus search again.

12. Dropout Detection Circuit

- (1). Black Band Dropout (BDO)

Dropouts are detected from the RF signal within IC-101. In order to stabilize the tracking servo against dropout, the BDO signal is supplied from IC-601 pin 37 to the phase difference limiter and used to hold the tracking servo operation.

- (2). Signal Dropout (SDO)

If an RF signal dropout is detected by the RF detector, or a BDO is detected, the SDO signal is supplied to the signal processor to stabilize the CLV servo operation.

The SDO signal is supplied to the signal processor to stabilize the CLV servo operation. The SDO signal, produced within IC-601, is used to stabilize PLL circuit operation.

13. Braking Circuit

The braking circuit applies a brake to the tracking actuator so as to speed the tracking servo pull-in sequence required after a SKIP or KICK operation.

The direction of actuator movement is detected by the OFTR DET (off track detection), CROSS (tracking cross) and DIRECTION signals from IC-102 and a brake signal is added to the tracking error signal.

16. Pin Description of AN8373S (IC1)

Pin No.	Mark	I/O Division	Function
1	AMP1	I	RF signal input (X30 amp.)
2	PDAD	I	Photo detector current input (A2)
3	PDA	I	Photo detector current input (A1)
4	PDBD	I	Photo detector current input (A4)
5	PDB	I	Photo detector current input (A3)
6	LPD	I	Non-inverting laser power input
7	LD	O	Laser power auto control output
8	FBL1	I	PD balance adjustment
9	FBL2	I	
10	TBL1	I	Tracking balance adjustment
11	TBL2	I	
12	FOOFS	I	Focus offset adjustment
13	IVA	O	Current/voltage conversion output (A)
14	IVB	O	Current/voltage conversion output (B)
15	FE	O	Focus gain adjustment output
16	FPI	I	Focus error signal input
17	TPI	I	Tracking error signal input
18	C. TPL	I	Tracking error filter capacitor input
19	C. TPH		
20	C. FPL	I	Focus error filter capacitor input
21	C. FPH		

Pin No.	Mark	I/O Division	Function
22	TPO	O	Tracking error signal output
23	FPO	O	Focus error signal output
24	FGC	I	Focus gain up signal input
25	TGC	I	Tracking gain up signal input
26	GD	I	Focus/tracking down signal input
27	PTO	O	Position detecting amp. output
28	PTI	I	Position detecting amp. input
29	PBO	O	Position detecting buffer output
30	POT	I	Position detecting buffer input
31	BDO	O	Dropout detection output
32	RFDET	O	RF detection signal output
33	SDO	O	Dropout detection pulse output
34	C. SBDO	I	Dropout detecting capacitor input
35	ARF	O	RF signal output
36	C. AGC	I	AGC detecting capacitor input
37	VCC	I	Power supply (+5 V input)
38	LDON	I	Laser power control input
39	RF IN	I	RF signal input
40	AMPO	O	RF signal output
41	VREF	O	Reference voltage output
42	GND	I	Ground terminal

17. Pin Description of AN8374S (IC601)

Pin No.	Mark	I/O Division	Function
1	LSA	I	Phase difference input (A)
2	LSB	I	Phase difference input (B)
3	TEOFS	I	Tracking offset adjustment
4	TE	O	Tracking gain adjustment
5	TEG	I	
6	TE OUT	O	Tracking error signal output
7	TE BPF	I	Tracking error gain detecting filter (Not used, open)
8	FEG	I	Focus gain adjustment
9	FE OUT	O	Focus error signal output
10	CLW	O	Triangular wave oscillator capacitor output
11	VREF	I	Reference voltage input
12	ARF	I	RF signal input
13	CDSL	I	Data slice filter capacitor input
14	FPC	I	Frequency difference signal input
15	GND	I	Ground terminal
16	C. PLL	I	PLL loop filter constant
17	VSS	I	Ground terminal
18	CLK	I	Frequency pull-in clock signal (88.2 kHz) input
19	SRF	O	Sliced and digitized RF signal output
20	PCK	O	Clock output extracted from SRF
21	EFM	O	EFM signal output synchronous with PCK

Pin No.	Mark	I/O Division	Function
22	VDD	I	Power supply (+5 V input)
23	SPCNT	O	Track crossing speed control output (Not used, open)
24	SENSE	O	Selector output (track crossing state)
25	TRV	O	Traverse servo control output
26	FLOCK	O	Focus lock signal output
27	KICK	O	Track kick signal output
28	LDON	O	Laser power control output
29	VDET	O	Focus/tracking gain up output (Not used, open)
30	CNT1	I	Control input (FOON: Focus servo ON signal)
31	CNT2	I	Control input (TRON: Tracking servo ON signal)
32	CNT3	I	Control input (KICKF: Kick direction (forward) command)
33	CNT4	I	Control input (KICKR: Kick direction (reverse) command)
34	TRVF	I	Traverse forward command signal
35	TRVR	I	Traverse backward command signal
36	RFDET	I	RF detection signal input
37	BDO	I	Dropout detection input
38	VCC	I	Power supply (+5 V input)
39	TVPO	O	Traverse position detecting resistor/capacitor inputs
40	TVPI	I	
41	BROUT	O	Tracking drive control output
42	BRIN	I	Tracking error signal input

18. Pin Description of AN8377 (IC602)

Pin No.	Mark	I/O Division	Function
1	PVCC	I	Driver power supply (+5 V input)
2	VCC	I	Power supply (+5 V input)
3	TB	O	External transistor base driving output
4	VMON	O	Voltage (+5 V) output
5	TVDI	I	Traverse error signal input
6	FDI	I	Focus error signal input
7	TDI	I	Tracking error signal input
8	VREF	I	Reference voltage input

Pin No.	Mark	I/O Division	Function
9	TD-	O	Inverting output of tracking driver
10	TD+	O	Non-inverting output of tracking driver
11	FD-	O	Inverting output of focus driver
12	FD+	O	Non-inverting output of focus driver
13	TVD-	O	Inverting output of traverse driver
14	TVD+	O	Non-inverting output of traverse driver
15	RESET	O	Reset signal output
16	PC	I	PC input (connect to GND)

1) Principle of the Digital Filter

The way an audio wave is converted to a digital signal, processed digitally, and then returned to a high-purity audio form, is more and more widely understood.

The conversion from analogue into digital signals is often called pulse code modulation, or PCM. The analogue waveform is first sampled (measured at regular intervals). Then it is quantized. This means that it is given the nearest numerical value in the large range of numbers used to transmit the digital sound information. The quantized value is then converted to binary form and encoded into a pulse train representing the series of numbers. This process of analogue to digital conversion is shown in figure 5-1.

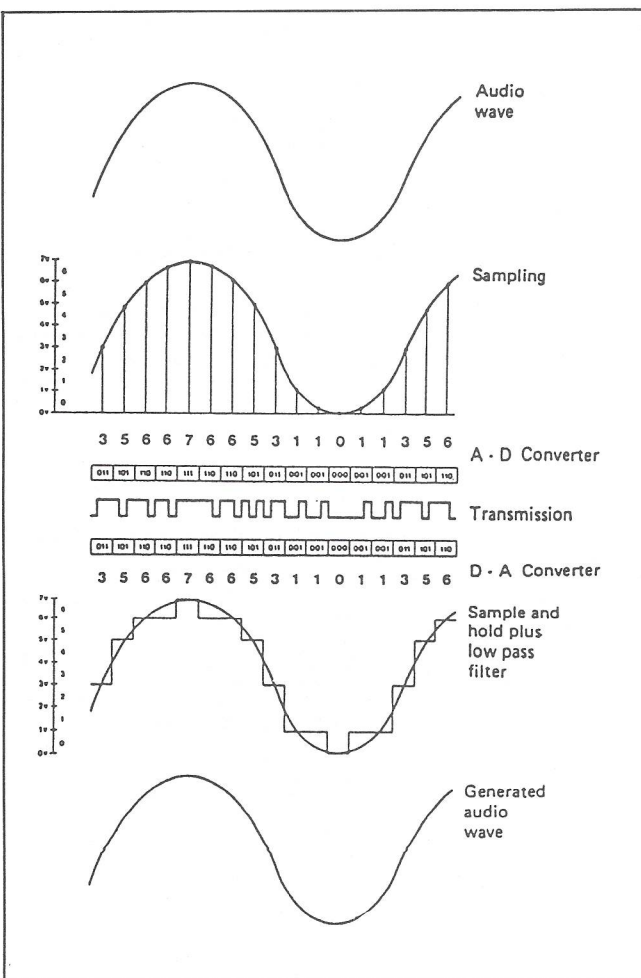


Figure 5-1

Figure 5-1 shows the digital sound reproduction process. The audio signal is sampled, quantized, converted to binary form, and encoded for recording or transmission.

Reversing the process produces a replica of the original signal. It is in this second stage that CD players use digital filtering to achieve exceptionally accurate reproduction.

The pulse train is fed into the transmission system (a cable or radio link, a tape, or a disc etc). At the reception or playback end, the pulse train is detected, decoded and regenerated in analogue form in the reverse process known as digital to analogue conversion.

2) How the Number of Bits Affects S/N

The accuracy of a digital system depends on the number of bits used for the binary number. This is because the number of bits determines the accuracy with which the samples taken from the original waveform can be quantized, i.e. expressed in digital form.

Two bits give four possible values for use in quantization, also three bits give eight possible values.

Every bit added doubles the number of possible values. In audio terms, this adds 6dB to the signal to noise ratio, because it reduces the quantization noise by a factor of 2.

The sixteen bits as used in the CD system give 65536 (2^{16}) possible values for quantization, and a signal to noise ratio of 96dB. The audio signals are encoded in the actual CD with this ratio.

96dB is approximation. An additional factor, which depends on the type of signal, is always present, regardless of the number of bits; for sine wave signals this makes the theoretical figure 97.8dB.

3) The Effect of the Sampling Frequency

Another important factor is sampling frequency, for efficiency in digital operations, this is kept low (44.1KHz)

at a little over twice the maximum audio frequency (20.0KHz). During the sampling process this frequency and its harmonics or multiples become modulated with the original waveform in a series of frequency bands as shows figure 19-2.

The bottom end of the lowest band is rather close to the top of the audio frequency range.

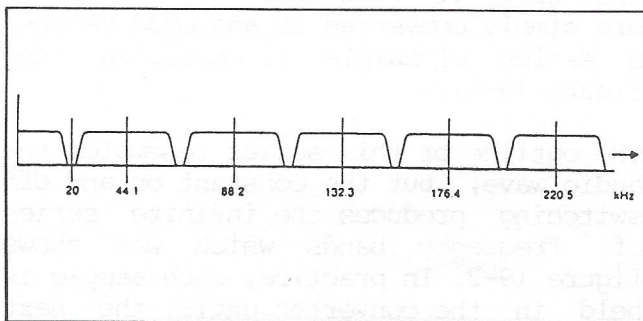


Figure 5-2

Figure 5-2 shows the frequency spectrum produced by sampling showing the periodic repetitions of the audio band. The bottom end of the lowest band is close to the audio frequency range.

4) Quantization Noise

In a digital audio system the audible noise has a different character to the familiar noise in an analogue audio system. The digital audio system background noise is called quantization noise.

The quantization noise of a system can be defined as the error signal between the original waveform before the analogue to digital conversion process, and the restored waveform after the digital to analogue conversion process. This error signal can be considered as both noise and distortion at the same time.

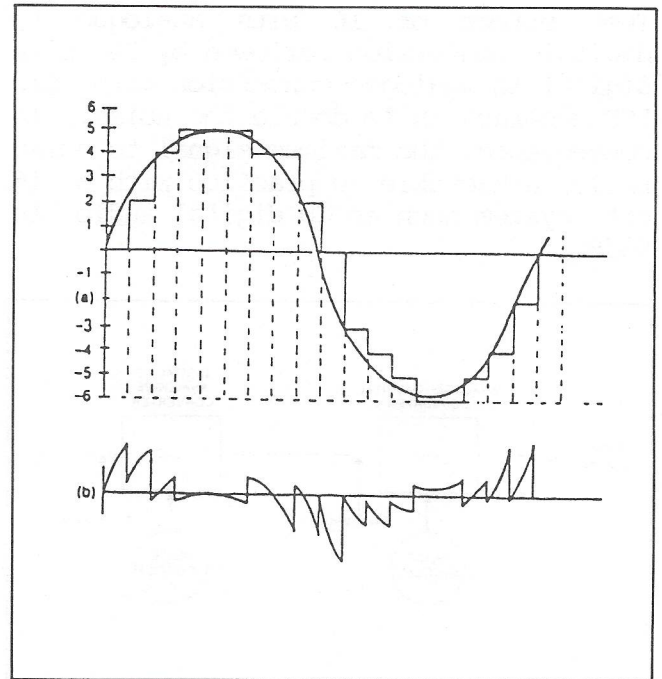


Figure 5-3

Figure 5-3 shows the quantization noise, The difference between the original audio waveform and the regenerated wave after D/A conversion (a) causes quantization noise (b).

The 16 bits analogue to digital conversion at the recording stage delivers a 16 bits digital signal with an intrinsic accuracy or signal to noise ration of maximum 96dB.

An ideal digital to analogue converter (D/A) would deliver an analogue output signal without adding noise to it. However, ideal D/A do not exist a small but finite electronic inaccuracy adds noise to the quantization noise of the digital signal.

The inaccuracy of a 16 bits D/A (the resolution limit of half the least significant bit) can be interpreted as a noise source that reduces the signal to noise ratio of the D/A to 96dB.

The effect of 16 bits analogue to digital conversion followed by 16 bits digital to analogue conversion with 1/2 LSB accuracy is to double the noise. In consequence, the maximum signal to noise ratio achievable in practice with a 16 bit system such as CD digital audio is 93dB.

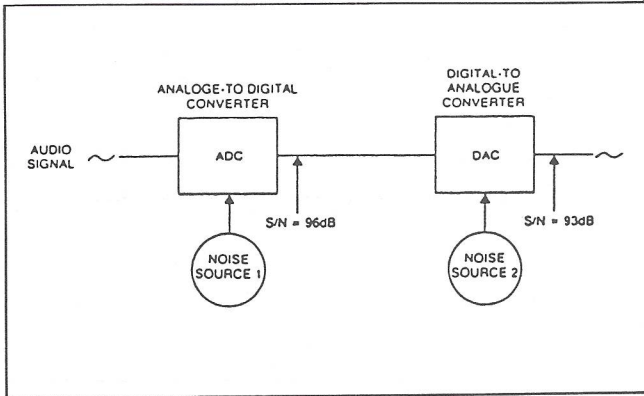


Figure 5-4

Figure 5-4 shows the signal to noise ratio deterioration in a digital chain, consisting of an A/D and a practical D/A.

5) Need for Filtering

With 16 bits (65536 possible numbers) and a sampling frequency of 44.1KHz, the CD system is theoretically capable of far higher reproduction quality than any conventional record player. But to achieve this quality, the output signal that the player derives from the disc must contain only the original audio frequencies, and not the periodic repetitions of the audio frequencies produced by the sampling (see figure 19-2).

These frequency bands can overload the power amplifier, and mix with audio frequencies or other signals to cause audible interference. They must be filtered out.

6) How D/A Conversion Operate

Figure 5-5 shows the generation process. Sampling produces only a series of sample (a) and hold (b) produces a step waveform closer to the original.

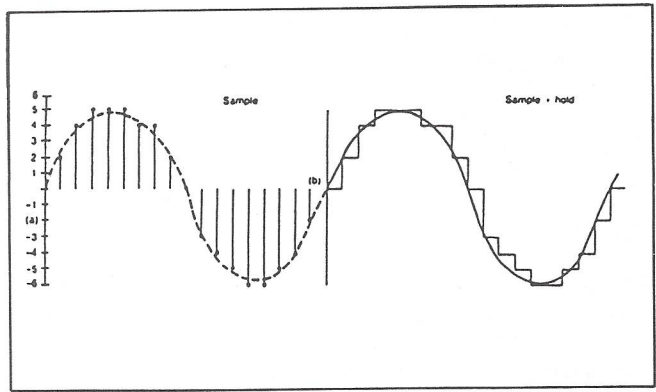


Figure 5-5

In the pulse train a series of binary number represents quantized samples of the original audio frequency. If these are simply converted to analogue values, a series of sample is produced (see figure 19-5).

The outline of this series resembles the audio wave, but the constant on and off switching produces the infinite series of frequency bands which was shown in figure 19-2. In practice, each sample is held in the converter until the next sample arrives, to produce a step waveform which is much nearer the shape of the audio waveform than the series of sample.

This sample and hold process is illustrated in figure 5-5 (b). It substantially reduces the frequency band components, as can be seen from figure 19-6. Significant components still remain, however particularly in the region near the audio band.

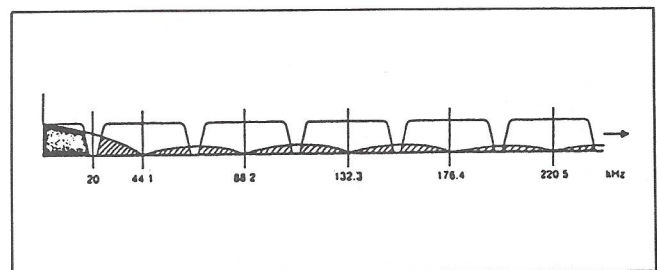


Figure 5-6

Figure 5-6 shows the effect of sample and hold. The remaining signal components are shown shaded.

7) Why Uses Digital Filtering

Most CD players suppress the frequency bands with conventional analogue filtering, after the signal has been converted back to analogue form. Such a filter is always complex, and can not avoid degrading the signal accuracy and introducing phase distortion.

Inevitable factors such as temperature changes and component aging affect performance. Most CD players on the

other hand, carry out this essential filtering process digitally, with no loss of accuracy, and phase distortion.

Digital filtering is a pure numerical calculation process with fixed coefficients. The fixing of the coefficients determines the filter characteristics which cannot afterwards vary.

8) Digital Filtering Process

A schematic diagram of the complete our D/A conversion system including digital filtering, is shows in figure 5-7.

It can be seen that our digital filtering uses the techniques of oversampling and noise shaping. Each plays a vital role in the digital filtering and conversion process.

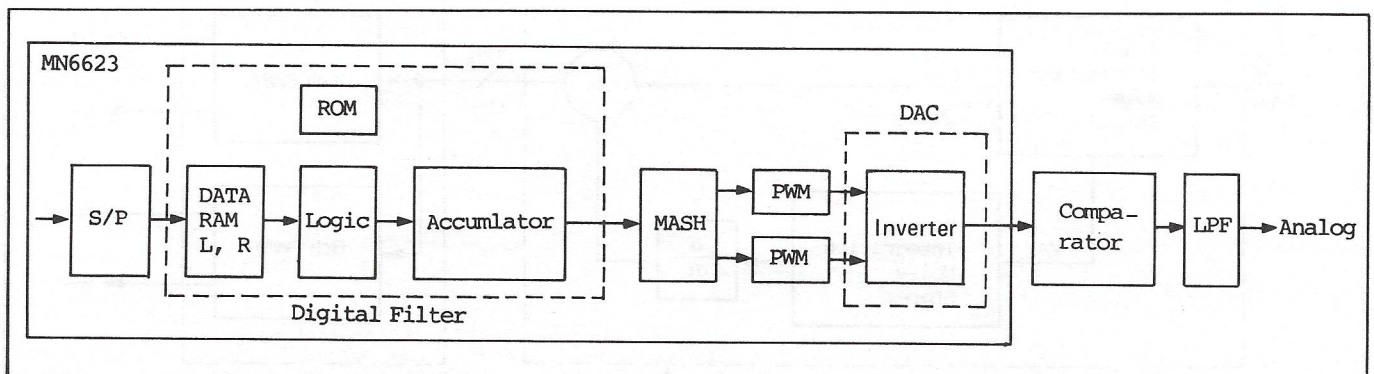


Figure 5-7

Oversampling:

Oversampling simply means multiplying the sampling frequency, in this case by four times. (44.1KHz becomes 176.4KHz.) The values of the intermediate samples are obtained by the calculation process

which determines the filter characteristic. An immediate effect of the oversampling as shows in figure 19-8 is to suppress all the lower frequency bands, centered on f_s , $2f_s$, and $3f_s$.

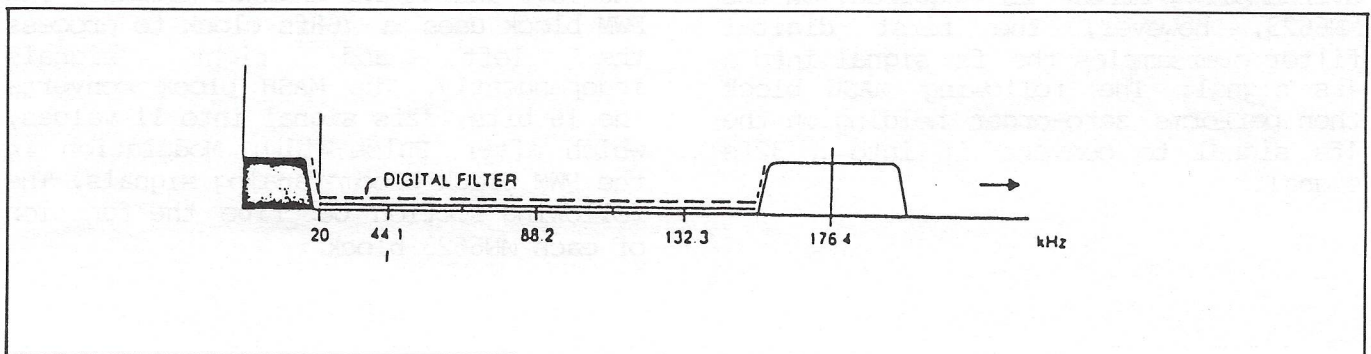


Figure 5-8

9) MN6623 Actual Operation

The MN6623 D/A converter system is composed of a digital filter, MASH, and PWM blocks as shown in figure 5-9.

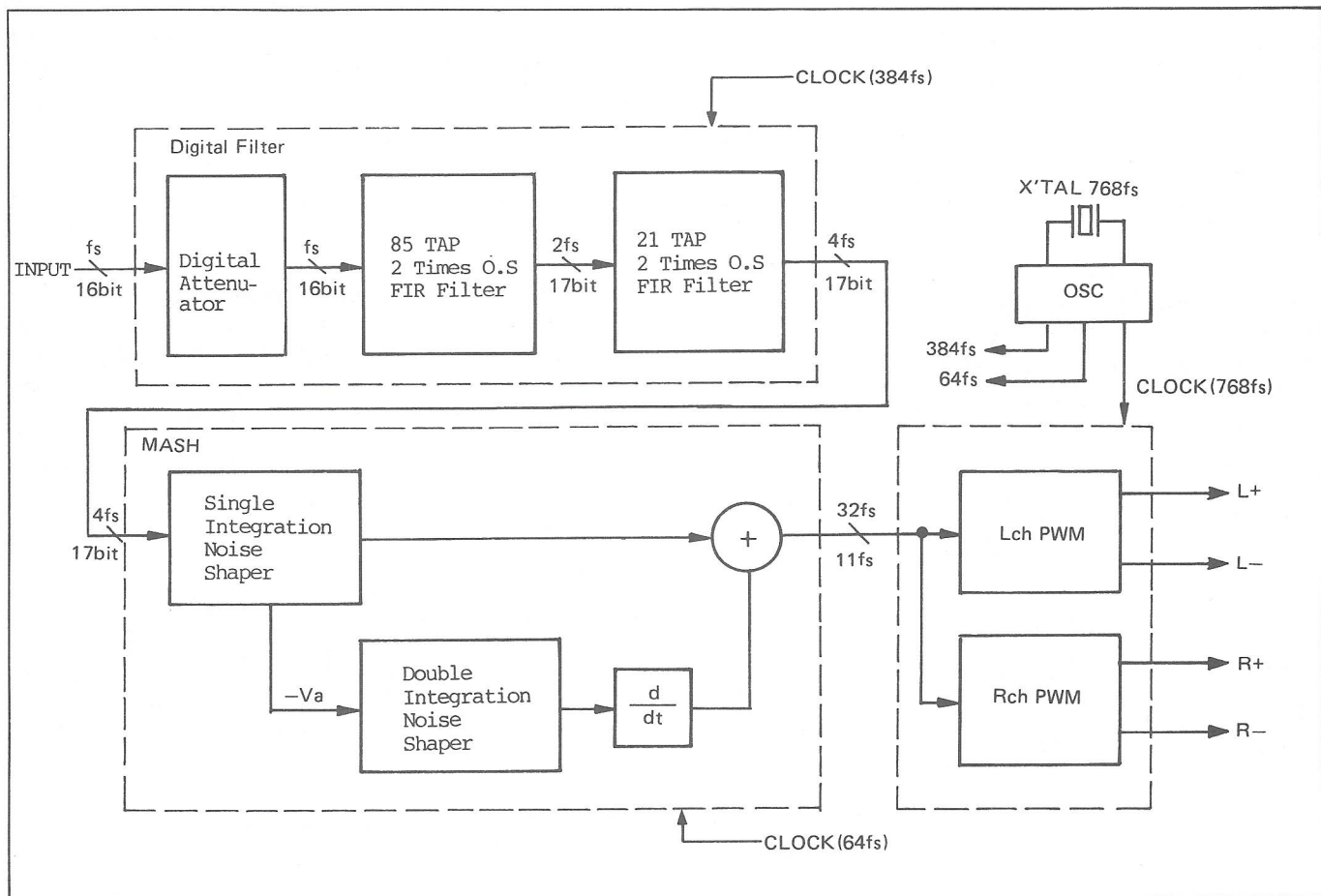


Figure 5-9 MN6623 Block Diagram

The MASH block operates at $32f_s$, where f_s is the sampled input signal frequency. This means that a 32 fold oversampling filter is required. On the MN6623, however, the first digital filter oversamples the f_s signal into a $4f_s$ signal. The following MASH block then performs zero-order holding on the $4f_s$ signal to convert it into a $32f_s$ signal.

The digital filter and MASH blocks use $384f_s$ and $64f_s$ clocks, respectively, to perform time-division multiplexing on the left and right channel signals. The PWM block uses a $768f_s$ clock to process the left and right signals independently. The MASH block converts the 16 bits, $32f_s$ signal into 11 values, which after Pulse-Width Modulation in the PWM block become analog signals. The following section describe the function of each MN6623 block.

Digital Filter Block

The digital filter block consists of three sub-blocks: ATT, FIR1, and FIR2 as shows in figure 5-10. It performs digital attenuation and oversampling to convert the input 16 bits, f_s data into a 16 bits, $4f_s$ signal.

The ATT attenuates the input data in a $1/64$ increment by multiplying the data by any 6 bits data. FIR1 and FIR2 are 2-fold oversampling digital filters with 85 and 21 taps each. Four-fold oversampling is accomplished by cascading the two -fold oversampling filters.

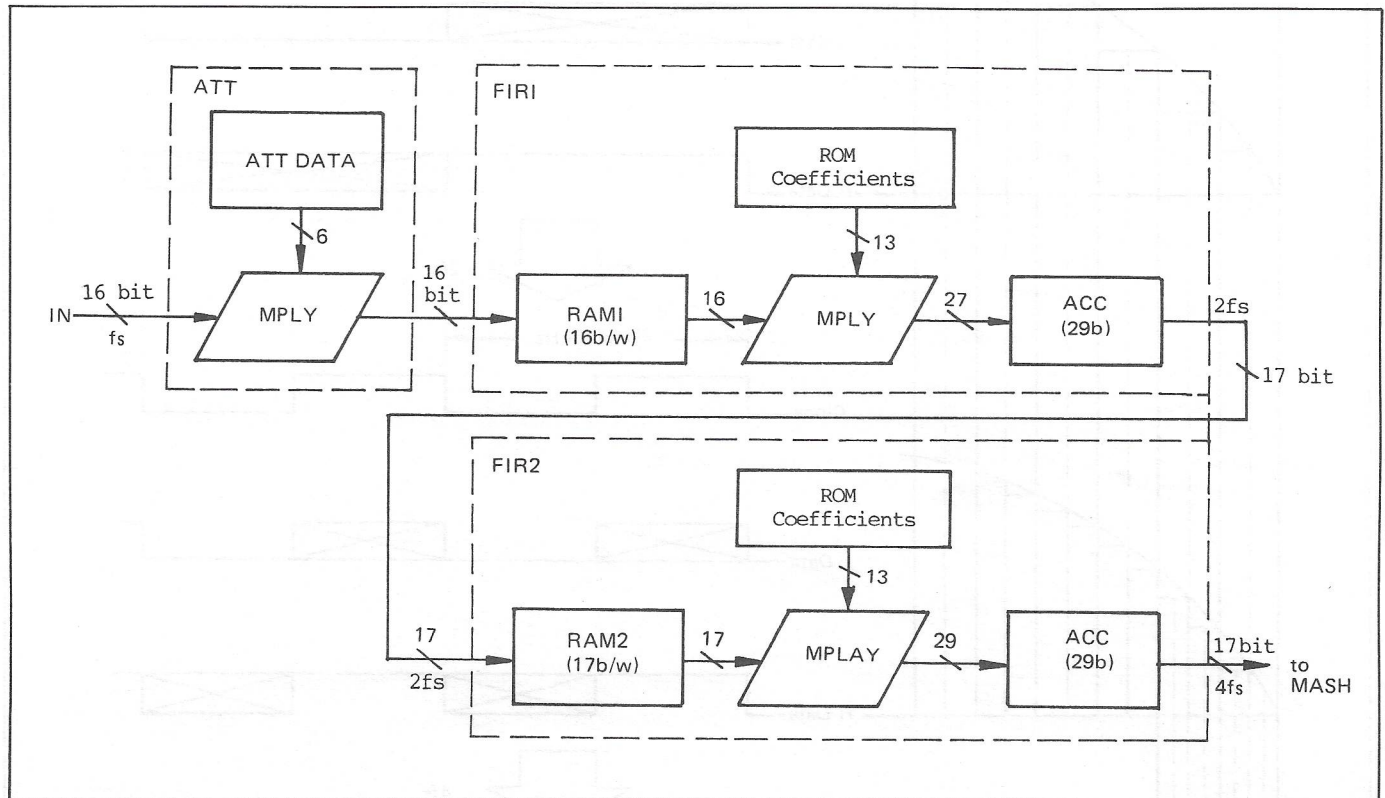
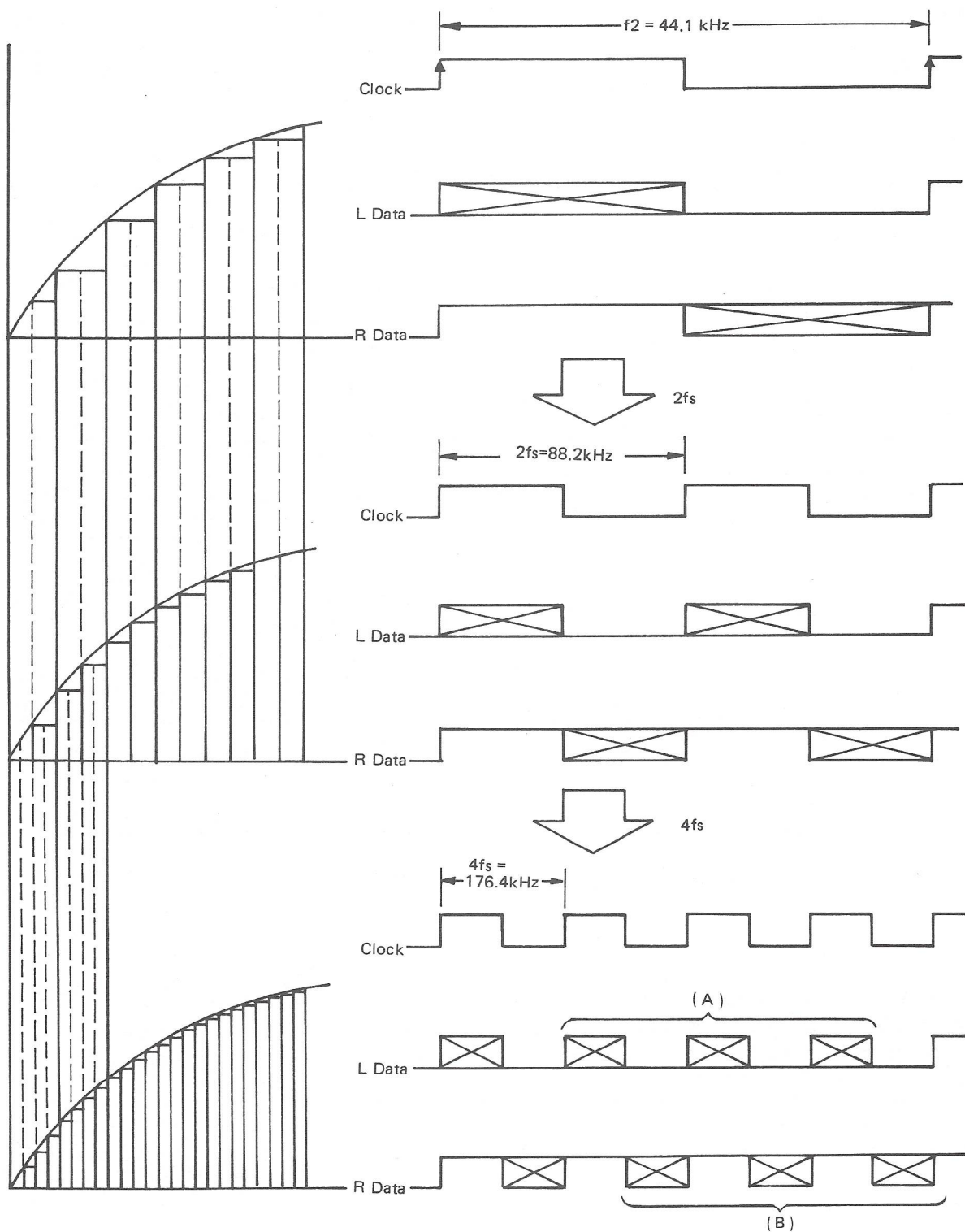


Figure 5-10 Digital Filter Block Diagram

The digital filter block contains many memories (delay logics), as shows in the block diagram above, with which it computes ROM coefficients at high speed while summing the coefficients to interpolate them. For more detail, shows in figure 5-11.



Note: (A) and (B) are the data created in the digital filter.

Figure 5-11

MASH Block

The MASH block consists of many noise shapers connected in series. The noise shapers include the signal and double integration types.

* Single integration noise shaper

The single integration noise shaper uses only a single integrator for noise shaping as shows in figure 5-12. The block marked "Q" in this figure is a local quantizer which adds quantization error to the input data. The relationship between output (Y) and input (X) for this type of noise shaper is:

$$Y = X + (1 - Z) \cdot V_q \dots\dots\dots (1)$$

The quantization error, V_q is a random value and the factor $(1 - Z)$ in this equation represents a differentiation respons. Eq. (1) tells us, therefore, that the spectrum of the quantization error, V_q in the single integration noise shaper has an attenuation slope of -6 dB/oct toward lower frequencies as shows in figure 5-14.

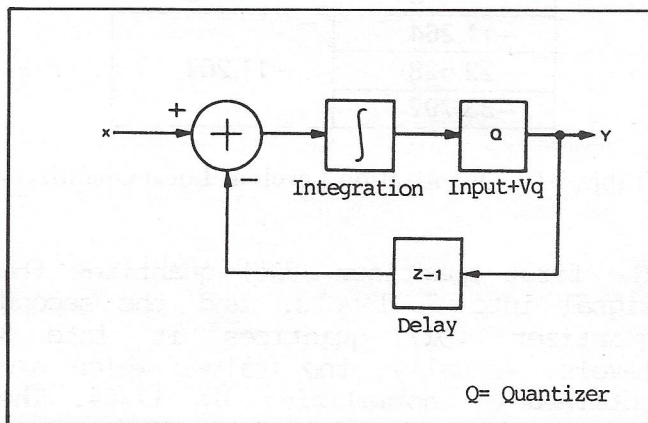


Figure 5-12 Single Integration Noise Shaper

* Double integration noise shaper

The double integration noise shaper uses two integrations in tandem as shows in figure 5-13. When the circuit is viewed at point W toward the output Y, it is

identical to the single integration noise shaper. Therefore, the relationship between Y and W is:

$$W = W + (1 - Z) \cdot V_q \dots\dots\dots (2)$$

W expressed as a function of X is:

$$W = \frac{1}{1 - Z} (X - Z \cdot Y) \dots\dots\dots (3)$$

From Eqs. (2) and (3) we obtain following overall output-input relationship:

$$Y = X + (1 - Z) \cdot V_q$$

If we compare this equation with Eq. (1), we find that the element $(1 - Z)$ is squared in the former equation.

This means that with the double integration noise shaper, the spectrum of the quantization error, V_q , has an attenuation slope of 12 dB/oct toward lower frequencies.

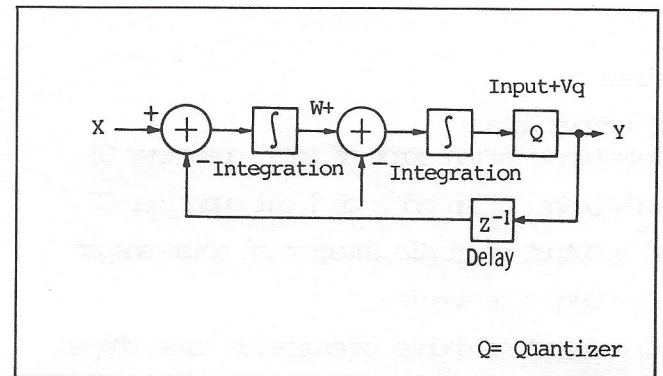


Figure 5-13 Double Integration Noise Shaper

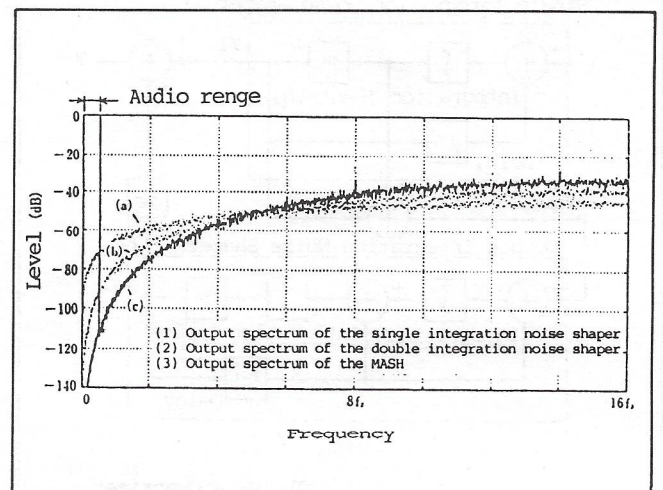


Figure 5-14 Noise Shaper Output Spectrum

The double integration noise shaper thus provides a greater attenuation of quantization noise than that obtained with the single integration type, as shows in figure 5-14.

The MASH block combines the single and double integration noise shapers in tandem so that the quantization error contained in the output of the first stage can be quantized again in the second stage further reduced.

Figure 5-15 is a block diagram of the MASH logic used on the MN6623 D/A converter. The MASH logic uses a single integration noise shaper in the first stage and a double integration noise shaper in the second stage. The outputs of the first and second stages of noise shaper, Y1 and Y2 are given by the following equations:

$$Y1 = X + (1 - Z) \cdot Vq1 \dots\dots\dots(5)$$

$$Y2 = -Vq1 + (1 - Z) \cdot Vq \dots\dots\dots(6)$$

Where

X = Input data

Vq1= Quantization error of local quantizer Q1

Vq2= Quantization error of local quantizer Q2

Y1 = Output of single integration noise shaper

Y2 = Output of single

Y2 = Output of double intergration noise shaper

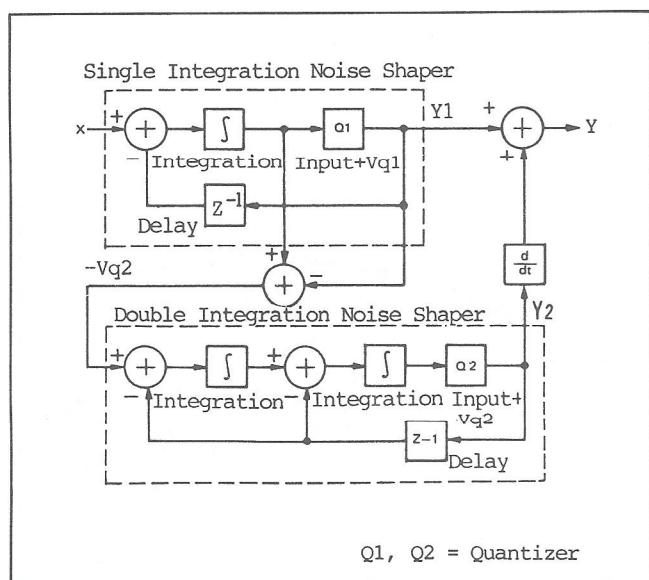


Figure 5-15 MASH Block Diagram

If we multiply both members of Eq.(5) by $(1 - Z)$ and add them to those of Eq. (6), we find that the quantization error is cancelled out.

Applying the output (Y2) of the second noise shaper to the differentiator and then adding the differentiator's output to the first noise shaper's output gives an overall MASH output, Y as represented by:

$$Y = X + (1 - Z) \cdot Vq2 \dots\dots\dots(7)$$

Eq.(7) indicates that the quantization error, Vq, is given in a third-order differentiation response (-18 dB/oct) , which gives further reduction in quantization noise

Table 5-1 lists the quantization levels at the first and second local quantizers, Q1 and Q2.

Q1 OUTPUT	Q2 OUTPUT
33,792	11,264
22,528	
11,264	
0	0
-11,264	-11,264
-22,528	
-33,792	

Table 5-1 Quantization Levels at Local Quantizers

The first quantizer (Q1) quantizes the signal into 7 levels. and the second quantizer (Q2) quantizes it into 3 levels. Actually, the values which are obtained by normalizing by 11264. The output values listed in Table 19-1 appear at Y1 and Y2. i.e. values +-3, +-2, +-1, and +-0 appear at Y1 and values +- and +-0 appear at Y2. Since the output of the second local quantizer (Q2) is differentiated and added to the output of the first local quantizer (Q1), output Y yields 11 different values.

A high output level and wide dynamic range are thus achieved by quantizing the 16 bit, 4fs data from the digital filter at a sampling frequency 8 times as high (32fs).

The following chart contains the timings of the 11 types of digital data yielded by the MASH block.

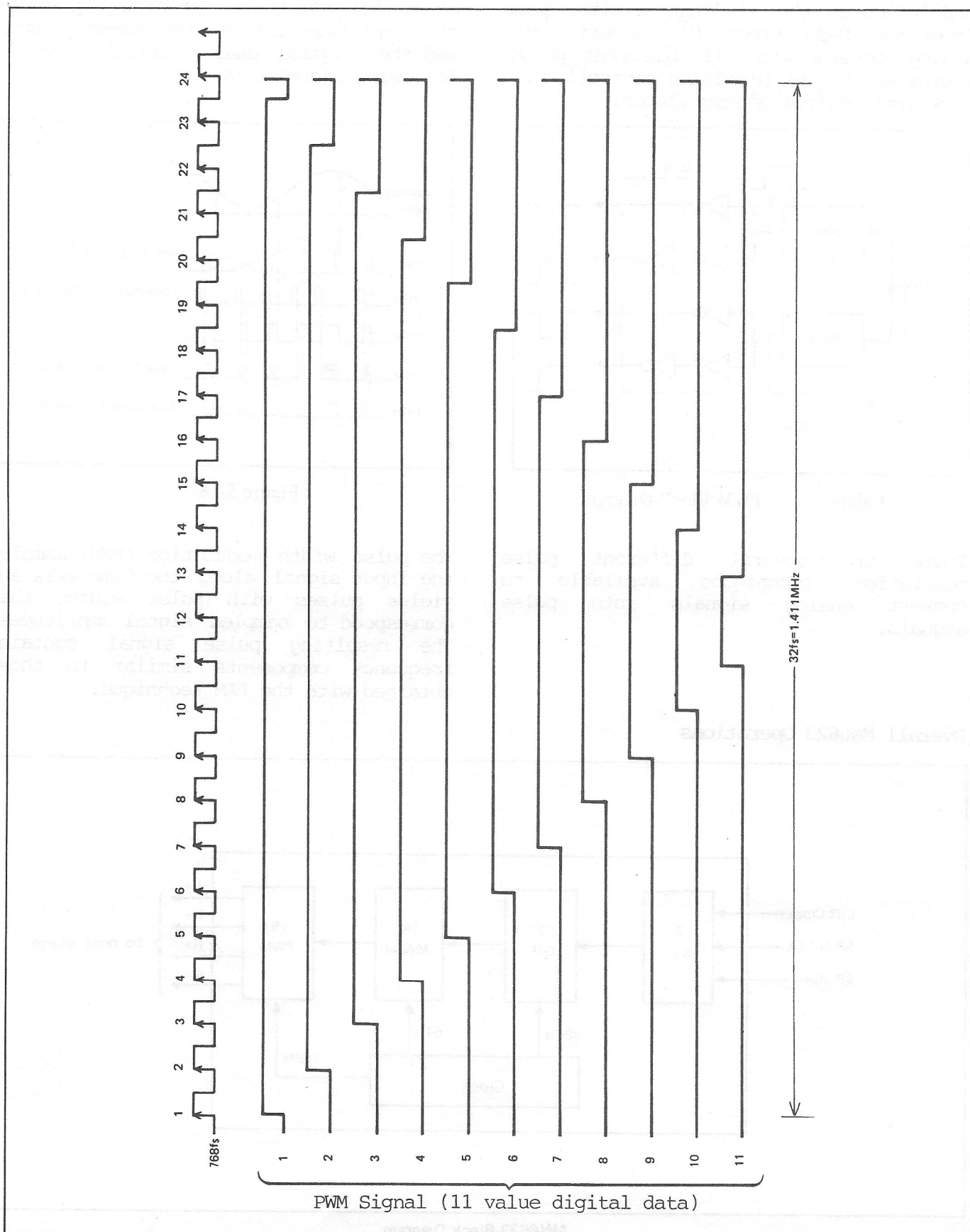


Figure 5-16 Output Data from MASH Part

PWM Block

The PWM block performs Pulse-Width Modulation on the 11 value digital data from the MASH block to convert into analog signals with 11 different pulse widths which are precision controlled by a crystal controlled oscillator.

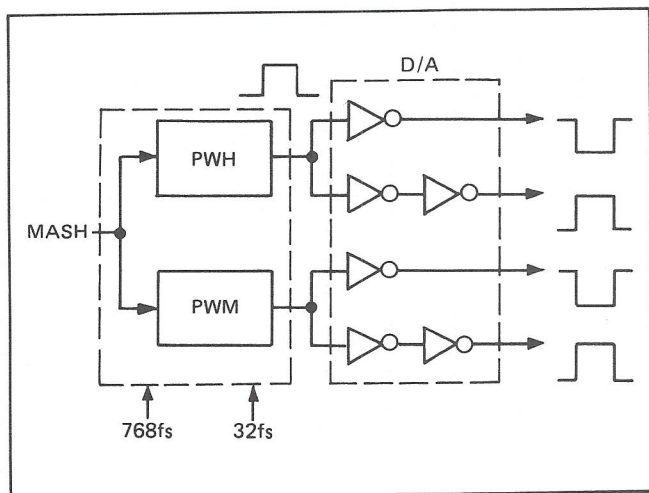


Figure 5-17 PWM Block Diagram

With the PAM, PWM, and PPM techniques, the signal is sampled along the time axis but is not quantized along the amplitude axis. All of these techniques preserve the amplitude information exactly as it was the original analog signal. (see the following figure 5-18)

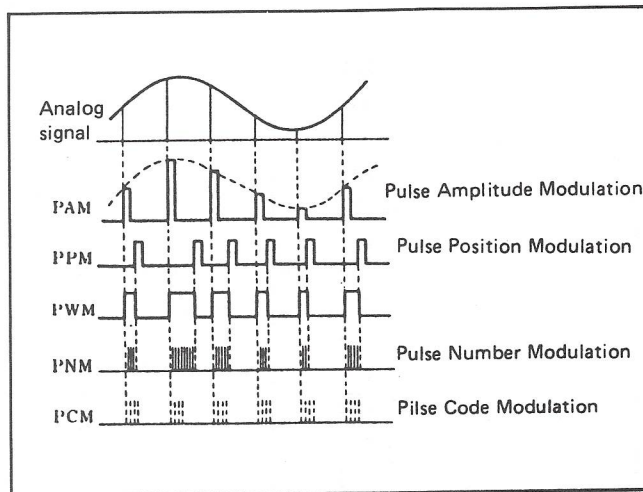
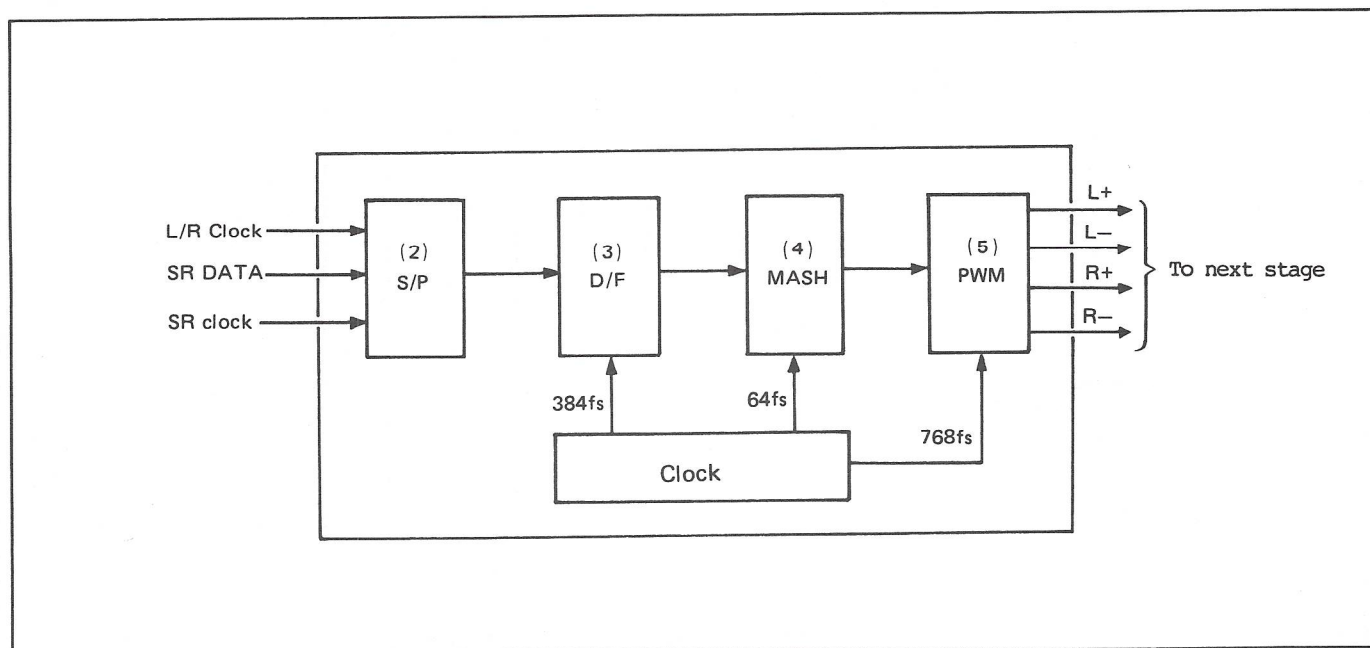


Figure 5-18

There are several different pulse modulation techniques available to convert analog signals into pulse signals.

The pulse width modulation (PWM) samples the input signal along the time axis and yields pulses with pulse widths that correspond to sampled signal amplitudes. The resulting pulse signal contains frequency components similar to those obtained with the PAM technique.

Overall MN6623 Operations



MN6623 Block Diagram